

FAB Group Project

Weekly Module 3: Representation

Objective | Define: (1) a set of **variables** and allowable **values**; extract all observations of variables from the course notes and from the text of your group project files that you have worked on so far; (2) a set of **constraints** on these variables; (3) a picture of your goal state as a **constraint network** that a student can use to propagate information about variables, similar to the experience of playing Sudoku.

Resources | (1) PDFs of Course Notes covered so far (Credit Assignment, Info. Proc., Search, Evolutionary Strategies; Representation); (2) A set of operators, heuristics, prompts, etc.; (3) (4) The definition of your goal state from last session.

Current gap | Currently-popular LLM tools support generating a single output in response to a single prompt. For example, to study the ideas in the course notes, you need to prompt an LLM for a text summary, or quiz, or diagram separately. Even if you use one prompt that asks the LLM to show you all of these outputs together, there is little explicit support for how to piece together the information in them. In the language of this week's session, there is little support for propagation, which limits the ability to reduce the combinatorial complexity involved in the problem of learning AI ideas. Put another way, whereas you may experience playing Sudoku as inherently supporting a rich and efficient propagation process, using an LLM requires you to do most of the propagation in your head. How can outputs of an AI system (LLM or otherwise) be decomposed into inherent, fundamental constraints that enable the magic of propagation?

Recommended Steps | I recommend proceeding in the following order:

0. Make sure you understand the concepts of constraints, constraint propagation, and constraint networks from the course notes on representation.
1. Refer to your evolutionary “schema” that you defined for the last session: building blocks of operators, heuristics, etc. structured into modules.
2. Discuss the actions a student might typically take based on these modules with respect to the problem-solving environment (i.e., the text of the course notes). What units of text do these actions most fundamentally concern? Keywords or tokens? Concepts? Phrases? Ultimately, these units of text should be the variables of your constraint network. Using an LLM, extract all of them (e.g., all concepts) from the course notes and all of your Group Project files.
3. Discuss how you can define constraints among one or more of these variables as simply and fundamentally as possible. For example, if your variables are concepts, the constraints could be that each concept may be defined in relation to other concepts at higher (more abstract) or lower (more concrete) levels, and in terms of relations such as causal structures. Define all the constraints.
4. Go back to your goal state definition from previous sessions. Now, based on your definitions of variables and constraints, draw a simple example of a goal state that a student might see using your system, depicted as a constraint network. Then, think about how a student may interact with this picture in ways similar to how they would interact with a Sudoku board when playing Sudoku — e.g., propagating information to make sense of how concepts (or other variables) underlying the AI ideas fit together.
6. Take a **screenshot** or photo of your constraint network, write a **short paragraph** describing what you have done, save as a **PDF**, and **upload on KLMS**.